



NATIONAL OCCUPATIONAL STANDARD FOR AN ANIMAL NUTRITIONIST

NOS.AN.01
First Edition

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 19th May 2022.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

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2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agricultural Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority (ZAQA)

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TABLE OF CONTENTS

FOREWORD	4
JUSTIFICATION	4
ACRONYMS AND ABBREVIATIONS	5
GLOSSARY OF TERMS	6
1. OVERVIEW	8
2. SCOPE.....	9
3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)	9
4. UNITS AND ELEMENTS.....	9
5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS	30
6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER.....	30
7. WORKING CONDITIONS/ENVIRONMENT	30
8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE	30
9. PHYSICAL DEMANDS ON THE BODY	31
ANNEX A.....	32
ANNEX B.....	33

FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to “***provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing***”.

Among other functions, ZAQA is responsible for “***determining national standards for any occupation***”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists.

JUSTIFICATION

Animal Nutritionists are the primary personnel responsible for ensuring that feed rations have optimum dietary requirements in accordance with species and production phase. This is important due to the fact that nutritional deficiencies and diseases may arise which can affect the health, well-being and productivity of the animals. In addition, animals with deficiencies are more susceptible to infectious diseases, as their immune systems do not operate optimally, whilst in a deficient state.

Therefore, the role of an Animal Nutritionist is to promote a better understanding of the effect of diet on the health, wellbeing, and productivity of animals. They use their scientific knowledge to advise on the effect of diet on animal health and wellbeing, and design and evaluate the diets of the animals in question. Proper nutrition gives animals the vigour to grow, develop, and reproduce, and strong immunity to fight off infections. All these advantages lead to more profitable and sustainable agriculture, hence, the need for the Animal Nutritionist at any viable animal farm enterprise.

ACRONYMS AND ABBREVIATIONS

AN	Animal Nutritionist
CS	Core Skill
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS for an animal nutritionist. It outlines the occupation, job description, purpose, ZQF level, minimum entry requirements, and sub sector among others.

NOS Code	NOS. AN.01
Occupation	Animal Nutritionist
Job Title	Animal Nutritionist
Job Description	Responsible for creating rations and balancing nutrition, to make sure that different animals' dietary needs are met.
Job Purpose	To provide basic ration formulations, based on mathematical calculations, for the mixing of animal stock-feeds, and, overseeing the mixing of rations according to the calculated rations, as well as working with the veterinary sector, to analyze nutritional disorders and provide advice on the necessary remedies.
ZQF Level	9
Sector	Agriculture
Sub sector	Livestock/Aquaculture
Other Economic Sector(s) in which the Occupation is Practiced	Livestock (animal clinics, dairy, wildlife) Education, (academia, research), Health (pharmaceutical Manufacturing (stockfeed production companies)
Other Similar Jobs that can be performed by an Animal Nutritionist	• consultants at farms • pharmaceutical companies • work for the government • animal nutrition lecturers
Minimum Educational Job Entry Qualification(s)	Master's degree in Animal Nutrition
Practicing License Requirements (if any)	Agricultural Institute of Zambia (AIZ)
Training/RPL	Bachelor's Degree in Animal Science, and/or Agriculture, with major in Animal Nutrition.
Minimum Job Entry Age	25 years
Prior Experience	Not applicable
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that an Animal Nutritionist must possess to be successful in his/her job role. It is applicable to Animal Nutritionists working in public, or private organisations, or self-employed.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Creativity
- Problem-solving skills
- Analytical skills
- Strong Mathematical and Biochemical skills
- Integrity and respect for confidentiality
- Interpersonal skills
- Commercial Awareness
- Attention to detail
- Ability to communicate effectively and clearly
- Self-motivated and a team worker
- Ability to plan and prioritize
- Quality consciousness
- Occupational health and safety oriented

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 6 Units representing the tasks that a job-holder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about Health, Safety and Environment Management].

Unit No.	01
Unit Title	Health, Safety and Environmental Management
Description	This unit is about maintaining safety, health and environmental protection for the individual, and for the stock-feed producing plant/farm.
Scope	This unit covers the following: • Health & Safety regulations. • Environmental protection.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & Safety regulations	To be competent, the individual must be able to: PC1. Read, interpret and implement national and organisational safety and health policies and regulations. PC 2. Assess risks and possible safety hazards of all aspects of operations
Environmental protection.	To be competent, the individual must be able to: PC3. Read, interpret and implement the environmental policies for the organisation PC4. Read, interpret and implement environmental standard operating procedures and policies of the organisation PC5. Read, interpret and implement national and global environmental regulations. PC6. Assess the Impact of animal farming systems on the environment, in particular energy use, emissions and greenhouse gases
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health and safety policies OK2. Company environmental policies OK3. Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Health and safety risks associated with mixing of different animal feeds. TK2. Personal Protective Equipment PPE is required for mixing different types of feeds.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory requirements for health & safety. RK2. Legislation relating to the animal feed industry, including feeding standards, and nutritional requirements for a variety of animal species
Skills (S)	

A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, and other company documents. CS5. Read from different sources- books, screens in machines, and signage. CS6. Understand the various colour codes, nomenclature, and acronyms related to the profession.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making

	The individual on the job must be able to: PS7 Recognize problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations, and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 2 [This unit is about Scientific Knowledge and Feed Production Competencies].

Unit No.	02
Unit Title	Scientific Knowledge and Feed Production Competencies
Description	This unit deals with knowledge and understanding of the scientific basis of animal nutrition and feed production competencies. Understanding nutritional requirements from the molecular through to the population level for either animal systems.
Scope	This unit covers the following: • Dietary requirements of different animal species at different stages of production cycles. • Nutritional breakdown of the key ingredients used in stock feeds. • Ingredients testing and formulation of balanced rations • Palatability and particle size considerations • Feed handling challenges
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Dietary Requirements of different animal species at different stages of production cycles	To be competent, the individual needs to be able to: - PC1. Determine the different key nutritional requirements of the different species. - PC2. Determine the nutritional demands over the production cycle of an animal (e.g. pregnancy, lactation, and different growth stages). - PC3. Analyse the nutritional requirements of the different species, in different production stages, in line with the ZABS.

	PC4. Analyse the anatomy, and physiology of a variety of animal species PC5. Analyse the feeding behaviour of animal species in relation to season, reproductive status, and competition for resources, and in relation to nutrient requirements, digestion, natural behaviour, and welfare.
Nutritional breakdown of the key ingredients used in stock feeds	To be competent, the individual needs to be able to: PC6. Establish the scientific basis and feed sources of nutrients PC7. Classify feed ingredients into energy, protein, vitamins and minerals, and additives. PC8. Demonstrate Knowledge of feed ingredients, their chemical composition, nutritive value, potential anti-nutritional factors, and toxins. PC9. Demonstrate, Knowledge of Biochemistry including the role of various pathways such as glycolytic and pentose phosphate pathways in biochemical reactions. PC10. Exhibit knowledge of required nutritional values for feed ingredients/ stock feed
Ingredients testing and formulation of balanced rations	To be competent, the individual needs to be able to: PC11. Demonstrate knowledge of sample collection techniques for different feed samples PC12. Use proximate analysis procedures or computerised animal feed analysis procedures to determine the nutritional composition of the key ingredients used in feed formulation. PC13. Evaluate the chemical and nutritional value of feeds, feed supplements, grass, and forage for commercial animals PC14. Demonstrate knowledge of manual (Pearson's square method) and - computer methods for ration formulation, statistics, and experimental design. PC15 Use of nutrient requirements to design practical feeding solutions for various categories of animals. PC16. Formulate diets and rations to maximize growth, reproduction, health, and performance PC17. Monitor feed formulations to meet quality performance and animal health standards PC18. Investigate human behaviour and perception and their influence on feed choice for animals PC19. Assess complementarity between the food needs of humans and farm animals PC20. Assess impact of animal farming systems on energy use and emissions of potential pollutants and greenhouse gases.

Palatability and particle size considerations	To be competent, the individual needs to be able to: PC21. Determine the required particle size, moisture, and hardness of the feed composition. PC22. Demonstrate the cost-benefit analysis of pelleting feed-in addressing palatability issues
Feed handling challenges	To be competent, the individual needs to be able to: PC23. Investigate nutritional disorders and the safe storage of ingredients and stock feeds PC24. Use the standards for moisture and particle size to address (out of feed occurrences) feed handling problems PC25. Demonstrate knowledge of feed contamination as a major risk associated with all feed handlers involved in the production process.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: • OK1. Company Quality, health and safety policies • OK2. Company environmental policies • OK3. Company regulations and global best practices • OK4. Company feed manufacturing processes including traceability and HACCP procedures
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Scientific basis and feed sources of nutrients TK2. Physiology of digestion and metabolism in nominated species TK3. Nutrient requirements for maintenance, production and performance in nominated species e.g. growth, lactation, egg production, wool production and sport TK4. Feeding and nutrition disorders of digestion and metabolism in animals TK5. Physiological and pathological effects of deficiencies (and excesses) of specific nutrients and food on a variety of animal species TK6. Complete and balanced nutrition versus supplementation and the impact on feed, performance and life-course of the animal TK7. Formulation of diets/feed products and management procedures for economical production, health and welfare in the food animals. TK8. Formulation of diets for non-agricultural animals eg riding horses, pets TK9. Pastures, fodder crop production and Range management TK10. Designing feeds to complement grass/forage in grazing animals e.g. horses and cattle.

	TK11. Designing feeding programmes that optimise the use of by-products from human food production, and lead to a net reduction in pollution and greenhouse gases TK12. Designing products/feeding programmes for specific clients/animals TK13. Operating machinery used to make feed and their limitations.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory agency requirements for feed manufacturing
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature, and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to:

	PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7 Recognize problems, and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 3 [This unit is about Food Chain and Quality Assurance].

Unit No.	03
Unit Title	Food Chain and Quality Assurance
Description	This unit is about the knowledge and understanding of the food chain and its impact on food choice. Integrating the food supply with dietary intake.
Scope	This unit covers the following: • Residue effects of dietary constituents for animals on human health and safety. • Animal feeding regimes and food quality assurance.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Residue effects of dietary constituents for animals on human health and safety	To be competent, the individual needs to be able to: PC1. Determine the effects of quantity and composition of dietary constituents for animals on the composition of food animals and their products. PC2. Determine the effects of different processing techniques on animal products for human consumption on nutrition and health in the human subject. PC3. Demonstrate knowledge of food chain.
Animal feeding regimes and food quality assurance	To be competent, the individual needs to be able to: PC4. Manipulate fat concentration and fatty acid composition in food animals. PC5. Detect and offer solutions to feeding management problems in the field. PC6. Appropriately label and use feedstuffs. PC7. Source and use appropriate ingredients for feedstuffs or for direct feeding. PC8. Educate dieticians, health professionals and the general public on the composition of foods of animal origin in relation to their possible effects on human health. PC9. Identify species differences in the composition of meat and milk, and effects (or non-effects) of different farming systems e.g. organic. PC10. Provide Advice on diet and feeding strategy to control e.g. stereotypic behaviour in horses, laboratory rodents, zoo animals, anxiety and aggression in dogs
Knowledge and Understanding (K)	

A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company feed manufacturing processes including traceability and HACCP procedures
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Feedstuffs and ingredients, their chemical composition, nutritive value, potential anti-nutrients and toxins TK2. Hazard Analysis and Critical control points (HACCP) and traceability mechanisms to ensure quality assurance and feed safety. TK3. The impact of animal feeding and management on risks of animal diseases TK4. Human health risks resulting from animal feeding practices, for example antibiotic resistance TK5. Effects of different processing techniques on animal products for human consumption on nutrition and health in the human subject
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Legislation for the control of the animal feed industry e.g. feed stuffs/marketing regulations and their certification or distribution permits
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussion. CS9. Respond appropriately to any queries.

	CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 4 [This unit is about Animal Health and Consultancy Services).

Unit No.	04
Unit Title	Animal Health and Consultancy Services.
Description	This unit is about understanding how to apply the scientific principles of nutrition for the promotion of animal health and provision of consultancy services; recognizing benefits and risks.
Scope	This unit covers the following: • Identification of symptoms experienced by animals with different types of nutrient deficiencies. • Nutrient deficiency Disorders, and recommendations for correction.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Identification of symptoms experienced by animals with different types of deficiencies	To be competent, the individual must be able to: PC1. Working with veterinary personnel, examine symptoms an animal is likely to display with different types of nutritional deficiencies. PC2. Diagnosis and control of mineral deficiencies in grazing animals. PC3. Develop dietary management plans to minimise the risk of disorders such as acidosis, ketosis, laminitis PC4. Practice Pasture management and parasite control, PC5. Control endemic infections and locomotor disorders
Nutrient deficiency Disorders, and recommendations for correction	To be competent, the individual must be able to: PC6. Identify feed types and samples to collect in order to test for nutritional deficiencies. PC7. Correctly advice the animal owner on how to correct the deficiency. PC8. Advising diets for (nominated) animals based on their individual requirements. PC9. Design formulation and administration of supplements to pasture Herd health planning Leaflets for clients including horse/pet owners PC10. Outline herd health programmes and feeding strategies to control deficiency disorders e.g 'leg weakness' in broiler chickens, post-weaning diarrhoea in piglets.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health and safety policies with regard to handling of feed and other samples. OK2. Company environmental policies with regard to handling of feed and other samples. OK3. Company regulations and global best practices with regard to the handling of feed and other samples.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of:

	TK1. Physiological and pathological effects of deficiencies (and excess) of specific nutrients TK2. Disorders of digestion and metabolism in animals attributable to feeding and nutrition TK3. The impact of animal feeding and management on risks of animal diseases, zoonotic diseases and other health risks to humans (e.g. antibiotic resistance). TK4. Use of nutrient requirements to design practical feeding solutions for (nominated) animals to nutritionally support maintenance, production, health and performance. TK5. Feeding behaviour of animal populations in relation to season, reproductive status and competition for resources TK6. Human behaviour and perception and how these influence feed choice for animals TK7. Trends in human health and nutrition and their impact on the consumption of food of animal origin
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements for health & safety RK2. Regulations regarding the minimum standards for stockfeeds (ZABS)
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual.

	CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems, and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 5 [This unit is about Stock Feed Manufacturing and Professional Conduct].

Unit No.	05
Unit Title	Stock Feed Manufacturing and Professional Conduct
Description	This unit is about the tasks that an animal nutritionist may need to complete when working for a company that formulates stock feed. Further, an understanding of professional conduct and the Association for Nutrition's Code of Ethics with evidence of good character
Scope	This unit covers the following: • Supervision of the stock-feeding mixing process. • Overseeing technology and equipment. • Engaging with clients, and promoting feed sales • Professional principles and standards
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Supervision of Stock-feed Mixing Process	To be competent, the individual must be able to: PC1. Supervise the mixing of feed ingredients according to a set ration formulae.
Overseeing Technology and Equipment	To be competent, the individual must be able to: PC2. Demonstrate feed-mixing machines, and engage with technicians to ensure optimal operations. PC3. Appreciate the computerised aspects of feed-mixing operations, and ensure that weights and measures are set correctly.
Engaging with clients, and promoting feed sales	To be competent, the individual must be able to: PC4. Engage with customers, and handle any queries and concerns. PC5. Attend events to promote the feed to customers. PC6. Work with the marketing team, to market the products. PC7. Provide clear and simplified information for the production of feed promotion brochures
Professional principles and standards	1. Competence and professional development in practice (Non-Maleficence)
	To be competent, the individual must be able to: PC8. Practice using an evidence-based approach within areas of competence, continuously develop and enhance expertise, and recognize limitations. PC9. Demonstrate in depth scientific knowledge of Animal Nutrition and behaviour. PC10. Assess the validity and applicability of scientific evidence without personal bias. PC11. Practice within the limits of their scope and collaborate with the inter-professional team.
	2. Integrity in personal and organisational behaviors and practices (Autonomy)

	To be competent, the individual must be able to: PC12. Comply with all applicable laws and regulations; including obtaining/maintaining a state license or certification if engaged in practice governed by animal nutrition statutes. PC13. Maintain and appropriately use credentials. PC14. Respect intellectual property rights, including citation and recognition of the ideas and work of others, regardless of the medium (e.g. written, oral, electronic). PC15. Provide accurate and truthful information in all communications.
	3. Professionalism (Beneficence) To be competent, the individual must be able to: PC16. Participate in and contribute to decisions that affect the well-being of patients/clients. PC17. Respect the values, rights, knowledge, and skills of colleagues and other professionals. PC18. Demonstrate respect, constructive dialogue, civility and professionalism in all communications, including social media. PC19. Refrain from communicating false, fraudulent, deceptive, misleading, disparaging or unfair statements or claims. PC20. Provide objective evaluations of performance for employees, co-workers, and students and candidates for employment, professional association memberships, awards, or scholarships, making all reasonable efforts to avoid bias in the professional evaluation of others PC21. Contribute to the advancement and competence of others, including colleagues, students, and the public.
	4. Social responsibility for local, regional, national, global nutrition and well-being (Justice) PC22. Collaborate with others to reduce health disparities and protect human rights. PC23. Promote fairness and objectivity with fair and equitable treatment. PC24. Contribute time and expertise to activities that promote respect, integrity, and competence of the profession. PC25. Promote the unique role of nutrition and dietetics practitioners. PC26. Engage in service that benefits the community in order to enhance the public's trust in the profession. PC27. Seek leadership opportunities in professional, community, and service organisations to enhance health and nutritional status while protecting the public.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard Operating Procedures for the mixing of feed. OK2. Standard Operating Procedures for health and safety. OK3. The production line. OK4. The Marketing Policy.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Feed manufacturing TK2. Understanding of biosecurity and hygiene. TK3. How to mix feed ingredients. TK4. How to demonstrate feed mixing machines, according to manufacturer's specifications. TK5. Customer Service and marketing.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Code of Ethics for the Nutrition and Dietetics Profession RK2. Regulations for stock feed standards.
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual.

	CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7 Recognize problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations, and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 6 [Extension Services, Training and Research in Animal Nutrition].

Unit No.	06
Unit Title	Extension Services, Training and Research in Animal Nutrition
Description	This unit is about extension services, training, and research in animal nutrition
Scope	This unit covers the following: • Provision of extension services to animal owners in nutrition • Training of animal science students • Conducting research in Animal Nutrition
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Provision of extension services to animal owners in nutrition	To be competent, the individual must be able to: PC1. Prepare presentations and training materials. PC2. Facilitate discussions with animal owners concerning the nutritional needs of animals and how they can economically meet these needs.
Training of animal science students	To be competent, the individual must be able to: PC3. Teach animal science students, who have an interest in nutrition. PC4. Supervise undergraduate students involved in animal nutrition research effectively and consistently PC5. Meet deadlines with regard to course coverage, marking of scripts, and practical assignments PC6. Work with professional boards and external examiners in his or her training currier PC7. Engage the industry, Zambia research institute, and other stakeholders in training needs assessments during curriculum reviews
Conducting research in Animal Nutrition	To be competent, the individual must be able to: PC8. Identify problems/ knowledge gaps in the animal nutrition industry PC9. Evaluate the chemical and nutritional value of feeds, feed supplements, grass and forage for commercial animals and pets. PC10. Conduct feed protocols and trials to maximise growth, reproduction, health and performance of animals. PC11. Research on the effectiveness of dietary regimes on growth rates and meat and animal product quality. PC12. Conduct animal-based studies and laboratory trials PC13. Support agricultural consultants in their work e.g. breeder specialists, veterinary officers etc. PC14. Interpret, apply, participate in and/or generate research to enhance practice, innovation, and discovery. PC15. Make evidence-based practice decisions, taking into account the unique values and circumstances of the patient/client and community, in combination with the practitioner's expertise and judgment.

Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. The training institutions Standard Operating Procedures. - OK2. Teaching methodology and administrative education policy - OK3. Academic professional Boards, affiliations and role of external examiners
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Nutritional needs of all species of animals, in different stages of their production cycles. - TK2. Biostatistics and experimental designs. - TK3. Research methodology and computer software packages used in data analysis. - TK4. Facilitation techniques and procedures in conducting seminars.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : - RK1. Regulatory Agriculture policy - RK2. National Development Plan and Vision 2030.
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: - CS1. Write in English, and give simple concise instructions. - CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: - CS3. Read and interpret internal/external documents. - CS4. Read and understand manuals, health and safety instructions, memos, other company documents. - CS5. Read from different sources- books, screens in machines, and signage. - CS6. Understand the various colour codes, nomenclature, and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: - CS7. Express statements or information clearly, so that others can hear and understand. - CS8. Participate in and understand the main points of discussions. - CS9. Respond appropriately to any queries. - CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques

	CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with the intent of satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to; Computer, Printer and relevant Software programs, calculator, Personal protective equipment (PPE) for fieldwork, and sampling containers (where required). Hammer mills, Mixers, pelleting machines, Oil expellers, soybean Extruders, solvent extractors, Moisture metres, stitching machines, NIR machines, laboratory equipment, platform scales, Kitchen scales.

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of an Animal Nutritionist include exposure to chemical (possible toxins), physical (injuries from live animals, and machinery) and biological (dust) hazards, as well as working in noisy dusty environments, with machinery, working night shifts etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include carrying out risk assessment and implementing appropriate control measures, organising workplace safety sensitization and awareness, supporting capacity building through training, adhering to company's safety and standard operating procedures at all times, paying attention to detail, consulting extensively within and outside one's department/team on safety and other issues.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include sitting and standing for long hours, sometimes areas that are noisy and dusty, and areas with limited lighting and ventilation. Working in confined spaces, working in dusty environments where feed is being mixed. For on-farm consultancy, the nutritionist can sometimes be exposed to increment weather, slippery conditions, and injuries from live animals.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Parties involved/interacting with the jobholder who are internal to the organisation include supervisors, subordinates, and other employees.

8.2 External/Outside the Organisation

Parties involved/interacting with the jobholder who are external include customers/clients, veterinary surgeons, government regulators, trainers, occupational health and safety associations, and academia etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to handle sitting and standing for long periods to time.
- If samples of feedstuffs are required, the nutritionist may need to bend, twist and reach out in order to collect the samples, and sometimes may need to carry heavy boxes containing samples.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NO will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.AN.01		
ZQF Level	9	Version Number	01
Sector	Agriculture	Date of Approval	19 th May 2022
Sub Sector	Livestock	Date of Last Review	N/A
Occupation	Animal Nutritionist	Date of Next Review	May 2027

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